Multi-Layered Random Forest-based Metaphoric Hand Gesture Interface in VR

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Abstract

We present a novel Natural User Interface (NUI) framework, called Meta-Gesture[1], for supporting intricate interactive scenarios in Virtual Reality. Meta-Gesture uses the gestures of holding and manipulating everyday tools commonly used in daily life. Specifically, the holding gesture is used to summon a virtual object into the palm of the hand, while the manipulating gesture is used to trigger the function of the summoned virtual tool. Our contributions include: (i) a novel NUI supporting a sophisticated level of interaction, such as a very small (nail-sized) VR object selection, which is achieved by combining a 3D palm pose estimator (publicly available) with a proposed Meta-Gesture estimator and (ii) a novel multi-layered random forest as a Meta-Gesture estimator, which classifies a static and dynamic gesture hierarchically, in a single classifier. We also present (iii) a novel voxel coding scheme called Layered Shape Pattern (LSP), which is configured by calculating the fill rate of point clouds (raw source of data) in each voxel on top of the palm pose estimation. We foresee numerous potential AR/VR applications of our system, which augment a user's experience (See Fig. 4).

1. Introduction

Using bare hands is the most natural way to interact with virtual objects when wearing HMD. Although there have been studies about bare hand-based gesture recognition [1, 4] in the field of AR/VR, the most recent methods cannot be directly repurposed to intricate interactive scenarios, which involves orientation-aware selection and manipulation of objects in wearable VR environment. They are insufficient due to the following challenges of the task: sophisticated level of interaction, variances of viewpoint. Especially when interacting with AR/VR objects in egocentric viewpoint, self-occlusion is the most challenging problem because the finger is bent under the back of the hand.

Addressing the above challenges, we propose Meta-Gesture, which performs simultaneous static and dynamic gesture estimation. The idea of Meta-Gesture was initiated from the following question; "What if we could magically summon a tool up to the hand and manipulate it as occasion demands, similarly to [2] but without any additional device?" Because we have already learned the gesture of grasping and manipulating each everyday tool, the gesture is intuitive and easy to memorise. Moreover, Meta-Gesture is also able to support a sophisticated level of interaction based on a summoned virtual tool, such as a pair of tweezers, rather than through direct bare hand contact with AR/VR devices for the interaction. Thus, interestingly, even though Meta-Gesture is independent of any additional devices, it has the same effect as utilising a physical tool.

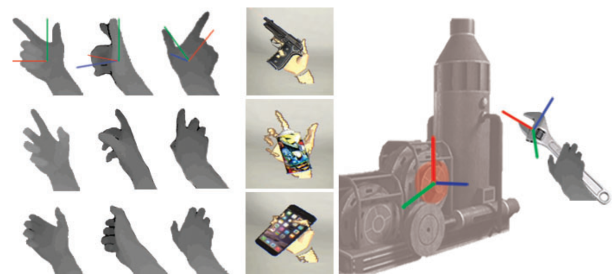


Figure 1. Our system supports orientation-aware selection and manipulation of small-sized AR objects by replicating the gestures of utilising a tool. More technically, Meta-Gesture identifies partial static 3D hand shapes for summoning a functional object up to the hand and then classifies 3D movement patterns of hand parts (i.e. functions) for manipulating the functional AR object [3].

The state-of-the-art computer vision (CV) methods, mainly categorized by discriminative [11, 10, 5, 8] and generative [7, 9] approaches, stably estimate 3D hand posture. Nevertheless, the methods are insufficient to be directly repurposed to Meta-Gesture, which requires the joint treatment of holding (static) and manipulating (dynamic) gestures, due to the following challenges of the task: imperfect feature representation in egocentric viewpoint, variances of static-dynamic parts for defining different gestures. Thus, the problem of static-dynamic gesture estimation cannot be straightforwardly solved with the conventional hand pose estimator.

[1]This paper is a summary version of [3].

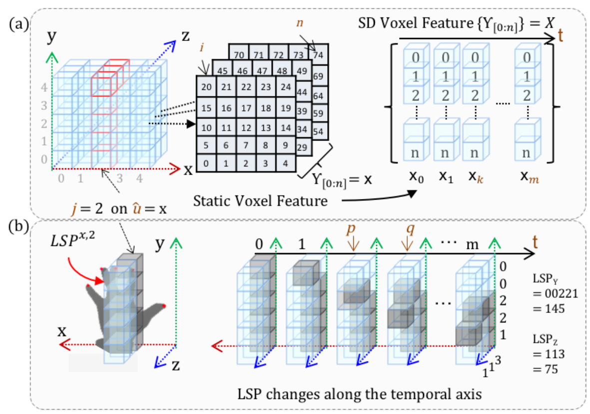


Figure 2. Example of Layered Shape Pattern (LSP) feature configuration: (a) example of the proposed Static-Dynamic (SD) voxel feature configuration (b) example of a layered shape on a selected voxel plane (changed through the sequence) and LSP pattern values calculated at the last frame based on octal number system.

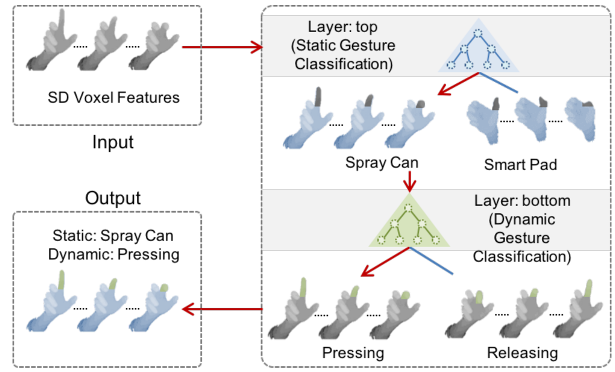


Figure 3. The proposed hierarchical Static-Dynamic (SD) gesture learning model.

2. Methodology

In this paper, we propose a multi-layered random forest utilising the proposed LSP feature [3]. The proposed forest is designed to take characteristics of both 3D partial static gesture (for retrieving a functional AR object) and 3D partial dynamic gesture status (for manipulating the AR object), with high stability in egocentric viewpoint. More specifically, the forest separately captures the different characteristics, such as the layer position showing the LSP pattern never changing through the temporal sequence and the discriminability between gestures or the tendency of the LSP value changing through the sequences at each node.

For that, the proposed forest learns from the sequence of Static-Dynamic (SD) voxel features consisting of hand point clouds, which are captured in the local coordinates of a hand. By voxelizing a cuboid region surrounding the hand, we were able to define the SD voxel feature based only on the position of point clouds without requiring any preliminarily image analysis. Moreover, the proposed forest is further able to define the best split function utilising the most important index of the feature and the offset time, which are both necessary considerations when estimating a 3D static gesture or a manipulating action status as a dynamic gesture at each split node.

2.1. Static-Dynamic (SD) Voxel Feature

Before making the SD voxel feature, as described in 3D Finger CAPE [4] utilising the state-of-the-art 3D palm pose estimator [6], we transform hand point clouds located in the camera coordinates into the local coordinates of the hand.

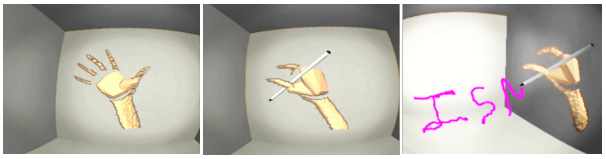
We propose the novel concept of LSP to capture a partial finger movement, which is technically a change in the amount of point clouds in a voxel, as well as static shape pattern on a layered voxel plane. In order to make a voxel plane $\mathrm{LSP}^{\hat{u},j}$, we selectively specify an axis-dependent index j along a designated axis $\hat{u}$ in the local hand coordinates as shown in Fig. 2(a). Based on the reconfigured voxel plane $\mathrm{LSP}^{\hat{u},j}$, two LSP_{axis} values can be extracted along each designated axis configuring a plane, as shown in Fig. 2(b). LSP_{axis} value changes between the two voxel planes, which are captured from two randomly selected different times, p and q, represent the same hand part (finger) movement, as shown in Fig. 2(b). Thus, the values along an axis representing the same dynamic gesture (finger movement) should correlate as they change, either both increasing or both decreasing, no matter the variances in viewpoint.

2.2. Multi-Layered Forest: SD Forest [3]

Our proposed forest is inspired by three different random forests: Transductive Regression Forests (TRF) [11], Multi-Layered Randomized Decision Forest (MLRDF) [5], and Spatio-Temporal Forest (STF) [4]. Specifically, hierarchical classifications for different tasks are inspired by TRF [11], the concept of holistic hand-shape classification is inspired by MLRDF [5], and spatio-temporal information learning in a process of interaction scenario is inspired by STF [4]. Based on each idea of prior works, we could structure the proposed novel type of random forest (Fig. 3), specialized for hierarchical static-dynamic gesture classification.

Overall, the proposed forest is composed of two layers representing an interdependent hierarchy, as shown in Fig. 3. Static gesture classification is first performed at the top levels. In this layer, multiple static gestures are classified based on the static part of a hand. As shown in Fig. 3, blue-colored hand parts represent static point clouds along a sequence of frames that are discriminable between different grabbing gestures, while gray-colored hand parts vary over the process of time. Once a static gesture is determined at the top levels, dynamic gesture (action status) is

(a)



(b)

(c)

Figure 4. Examples of Meta-Gesture use cases: (a) in an elaborate AR object manufacturing (assembling) scenario contributing to select a thin and small component in a cluttered scene; (b) with in-air pen annotation in 3D space with split letters; (c) in immersive gaming, summoning different types of weapon, such as an AR gun or an AR sword, intuitively.

classified by checking the learnt pattern of partial finger movements along the timeline. Green-colored hand parts obviously produce different patterns along the sequence of frames because gray-colored hand parts are static for a grabbing shape. Then, finally, static and dynamic gestures are classified through the layers of the proposed forests.

3. Conclusions

This paper presents Meta-Gesture conducting simultaneous static and dynamic gesture estimation under various 3D functional object manipulation scenarios in wearable AR/VR, as shown in Fig. 4. For supporting this joint static-dynamic gesture estimation, we proposed a novel voxel coding scheme called LSP and a novel multi-layered random forest. We found that the proposed Meta-Gesture encourages more intuitive interactions in the scenarios of the summoned function-equipped AR object manipulation, which is directly extendable into AR environment and independent of rotation and translation of the hand. In the future work, we plan to extend this work for more enhanced and augmented Natural User Interface (NUI), including work utilising a real object and different combinations of holding and finger gestures, and expand the static and dynamic gesture estimation from function-equipped AR objects to giving additional functionalities to real objects.

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